NMAMIT, NITTE

DEPARTMENT OF MCA

III SEM MCA

TASK-1

Subject: REACT NATIVE AND KOTLIN FOR MOBILE APPLICATION DEVELOPMENT

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1. Explain the concept of Shared Preferences and discuss its applications in Android.

Shared Preferences is a simple mechanism used in Android to **store and retrieve small amounts of data** persistently in the form of **key-value pairs**. The stored data remains available even after the application is closed and reopened.

Applications of Shared Preferences

1. **Storing user preferences:**
   Used to save settings such as theme selection, language, font size, etc.
2. **Remembering login information:**
   Can store simple information such as whether a user is logged in.
3. **Saving application settings:**
   Used for storing options selected by the user, such as notification settings.
4. **Storing small amounts of data:**
   Suitable for simple values such as strings, integers, and Boolean values.
5. **Maintaining application state:**
   Helps remember certain information between different sessions of the application.

Example

For a login application, Shared Preferences can store:

"isLoggedIn" → true

When the user opens the app again, the application can read this value and determine whether the user has already logged in.

In short: Shared Preferences are mainly used for **persistent storage of small, simple key-value data**, especially user preferences and application settings. The provided material identifies Android development as using platform-specific Android tools and technologies, but it does not specifically detail Shared Preferences.

2. Compare Internal Storage and External Storage based on security, accessibility, advantages, and limitations.

Basis	Internal Storage	External Storage
Security	More secure because data is private to the application.	Less secure because files may be accessible to other apps/users.
Accessibility	Mainly accessible only by the application.	Can be accessed more easily by other applications and the user.
Advantages	Provides better security and reliable storage for app-specific data.	Useful for storing larger files such as images, videos, and documents.
Limitations	Limited space and data is removed when the app is uninstalled.	Less secure and may not always be available, especially with removable storage.

In short: Internal storage is preferred for **private application data**, while external storage is suitable for **large files that may need broader accessibility**.

3. Explain the architecture of SQLite Database and discuss its features.

SQLite is a lightweight, embedded relational database used in Android applications to store structured data locally.

Architecture of SQLite

SQLite follows a simple **embedded architecture**:

Application → SQLite API → SQL Compiler → Storage Engine → Database File

- **Application:** Sends SQL commands such as CREATE, INSERT, UPDATE, and SELECT.
- **SQL Compiler:** Parses and converts SQL statements into instructions.
- **Storage Engine:** Executes these instructions and manages data in tables.

- **Database File:** Stores the actual database on the device.

Features of SQLite

1. **Lightweight:** Requires very little memory and system resources.
2. **Serverless:** No separate database server is required.
3. **Self-contained:** The complete database is stored in a single file.
4. **SQL Support:** Supports standard SQL operations for managing data.
5. **Transactional:** Provides reliable transactions using ACID properties.
6. **Open Source:** SQLite is freely available and widely used.

In short: SQLite provides a **simple, lightweight, serverless, and reliable** way to store structured data locally in Android applications.

4. Explain CRUD operations in SQLite Database with suitable examples.

CRUD stands for **Create, Read, Update, and Delete**. These are the basic operations used to manage data in an SQLite database.

1. **Create (INSERT):** Adds new records to the database.
   INSERT INTO Student VALUES (1, 'Rahul', 20);
2. **Read (SELECT):** Retrieves data from the database.
   SELECT * FROM Student;
3. **Update (UPDATE):** Modifies existing records.
   UPDATE Student SET age = 21 WHERE id = 1;
4. **Delete (DELETE):** Removes records from the database.
   DELETE FROM Student WHERE id = 1;

In short: CRUD operations allow an Android application to **add, retrieve, modify, and remove** data stored in an SQLite database.

5. Analyze the differences between Shared Preferences, Internal Storage, External Storage, and SQLite Database.

Feature	Shared Preferences	Internal Storage	External Storage	SQLite Database
Purpose	Small key-value data	Private app files	Large files	Structured data
Security	Relatively secure	Highly secure/private	Less secure	Secure app-level database

Accessibility	Mainly by the app	Only by the app	More accessible	Mainly through the app
Advantages	Simple and easy to use	Good for private data	Suitable for large files	Supports SQL and complex data
Limitations	Not suitable for large/complex data	Limited app-specific space	Security and availability issues	More complex than simple storage

In short: Shared Preferences is best for **small settings**, Internal Storage for **private files**, External Storage for **large accessible files**, and SQLite for **structured and relational data**.

6.Explain Android Notification architecture and discuss the role of sound, vibration, and LED lights.

A **notification** is a message displayed by Android to inform the user about an event, update, or activity in an application.

Notification Architecture

The basic flow is:

Application → Notification Manager → Android System → Notification → User

1. **Application:** Creates a notification containing information such as title and message.
2. **Notification Manager:** Handles and sends the notification to the Android system.
3. **Android System:** Displays the notification according to the user's settings.
4. **User:** Receives and interacts with the notification.

Role of Sound, Vibration and LED Lights

- **Sound:** Produces an alert tone to notify the user when a notification arrives.
- **Vibration:** Provides physical feedback, useful when the phone is in silent mode.
- **LED Light:** On supported devices, a flashing LED provides a visual indication of a new notification.

In short: Notifications use **sound, vibration, and LED lights** as different ways to attract the user's attention.

7. Discuss the process of Image Capture and Video Recording in Android applications.

Android applications can use the **device camera** to capture images and record videos.

Process

1. **Request Camera Permission:**
   The application must request the required camera permission from the user.
2. **Open Camera:**
   The app uses Android camera APIs or a camera intent to access the device camera.
3. **Capture Image:**
   The user takes a photo, and the captured image is saved to the specified storage location.
4. **Record Video:**
   The camera is switched to video mode. The user starts and stops recording, and the video is saved as a media file.
5. **Access the Media:**
   The application can display, process, or share the captured image or recorded video.

In short: The process involves **getting permission → accessing the camera → capturing/recording → saving the media → using or sharing it**.

8. Explain the MediaPlayer class and discuss its multimedia capabilities in Android.

The **MediaPlayer** class is an Android API used to **play audio and video media** in applications. It manages the media playback process from loading the media source to controlling its playback.

Multimedia Capabilities

1. **Audio Playback:** It can play music, sound effects, and other audio files.
2. **Video Playback:** It supports playback of video media in an application.
3. **Playback Control:** Provides operations such as **start, pause, stop, and seek** to control media.
4. **Media Sources:** Media can be loaded from **local files or remote sources**, depending on the supported format and access.

5. **Playback Management:** It handles different stages of media playback, allowing the application to control and monitor the media.

Example: A music application can use MediaPlayer to load a song, start playing it, pause it, and resume it when required.

In short: MediaPlayer provides Android applications with a convenient way to **play and control audio and video**, making it useful for music players, video applications, games, and educational apps.

9. Evaluate different Android storage mechanisms based on performance, security, and usability.

Android provides different storage mechanisms depending on the **type and amount of data** that needs to be stored. SQLite is also included in the Android platform stack.

Storage	Performance	Security	Usability
Shared Preferences	Fast for small data	Relatively secure	Very easy; best for key-value settings
Internal Storage	Good for app-specific files	High security and privacy	Easy, but mainly accessible by the app
External Storage	Good for large files	Less secure	Useful for files that need broader access
SQLite Database	Efficient for structured data and queries	Good for app-managed data	Suitable for large and complex structured data

Conclusion

Shared Preferences is best for small settings, **Internal Storage** for private app files, **External Storage** for large accessible files, and **SQLite** for structured and relational data.

10. Analyze the importance of Notifications, Camera, MediaPlayer, and SQLite in developing modern Android applications.

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